

حالات استخدام sandwich theorem :-
Sec. 2.2

[1] إذا كانت الدوال على شكل

$$f(x) \leq g(x) \leq h(x)$$

ولمطب limit الدالة الو $\rightarrow$

[2] عند وجود $\sin(\pm\infty)$ أو $\cos(\pm\infty)$
بعد التعويض المباشر

$\leftarrow$ في حالة كانت n عدد (odd):

$$-1 \leq \sin^n(\cdot) \leq 1 \quad / \quad -1 \leq \cos^n(\cdot) \leq 1$$

$\leftarrow$ في حالة كانت n عدد (even):

$$0 \leq \sin^n(\cdot) \leq 1 \quad / \quad 0 \leq \cos^n(\cdot) \leq 1$$

★ ال $\sin$ وال $\cos$ دائماً تعطي نواتج بين $[-1, 1]$.

$$(a^x)' = a^x \ln a$$

$$(\sin^{-1} x)' = \frac{1}{\sqrt{1-x^2}}$$

$$(\sin x)' = \cos x$$

$$(\cos x)' = -\sin x$$

$$(\cos^{-1} x)' = \frac{-1}{\sqrt{1-x^2}}$$

$$(\tan x)' = \sec^2 x$$

$$(\cot x)' = -\csc^2 x$$

$$(\tan^{-1} x)' = \frac{1}{1+x^2}$$

$$(\sec x)' = \sec x \tan x$$

$$(\csc x)' = -\csc x \cot x$$

$$(\cot^{-1} x)' = \frac{-1}{1+x^2}$$

$$(e^{u(x)})' = e^{u(x)} \cdot u'(x)$$

$$(a^{u(x)})' = a^{u(x)} \cdot \ln a \cdot u'(x)$$

$$(\sec^{-1} x)' = \frac{1}{x\sqrt{x^2-1}}$$

$$(\ln(u(x)))' = \frac{u'(x)}{u(x)}$$

$$(\csc^{-1} x)' = \frac{-1}{x\sqrt{x^2-1}}$$

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

$$\tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$(\sinh x)' = \cosh x$$

$$(\csc h x)' = -\csc h x \coth x$$

$$(\cosh x)' = \sinh x$$

$$(\operatorname{sech} x)' = -\operatorname{sech} x \tanh x$$

$$(\tanh x)' = \operatorname{sech}^2 x$$

$$(\coth x)' = -\operatorname{csch}^2 x$$

$$\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = \lim_{\theta \rightarrow 0} \frac{\theta}{\sin \theta} = 1$$

$$\lim_{\theta \rightarrow 0} \frac{\tan \theta}{\theta} = \lim_{\theta \rightarrow 0} \frac{\theta}{\tan \theta} = 1$$

$$\lim_{\theta \rightarrow 0} \frac{\cos \theta - 1}{\theta} = \lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\theta} = 0$$

$$\sin x^4 = \cos x^4 \cdot 4x^3$$

$$\sin^3 x = 3 \sin^2 x \cdot \cos x$$

$$\sin^2 \sqrt{x} = 2 \sin \sqrt{x} \cdot \cos \sqrt{x} \cdot \frac{1}{2\sqrt{x}}$$

$$(\log_a x)' = \frac{1}{x \ln a}$$

$$\int x^n dx = \frac{x^{n+1}}{n+1} \quad (n \neq -1)$$

$$\int e^x dx = e^x$$

$$\int \sin x dx = -\cos x$$

$$\int \sec^2 x dx = \tan x$$

$$\int \sec x dx = \ln |\sec x + \tan x|$$

$$\int \tan x dx = \ln |\sec x|$$

$$\int \sinh x dx = \cosh x$$

$$\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right)$$

$$\int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \ln \left| \frac{x-a}{x+a} \right|$$

$$\int \sec x \tan x dx = \sec x$$

$$\int \frac{dx}{a^2 - x^2} = \frac{1}{2a} \ln \left| \frac{x+a}{x-a} \right|$$

$$\int \frac{1}{x} dx = \ln |x|$$

$$\int b^x dx = \frac{b^x}{\ln b}$$

$$\int \cos x dx = \sin x$$

$$\int \csc^2 x dx = -\cot x$$

$$\int \csc x \cot x dx = -\csc x$$

$$\int \csc x dx = \ln |\csc x - \cot x|$$

$$\int \cot x dx = \ln |\sin x|$$

$$\int \cosh x dx = \sinh x$$

$$\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} \left(\frac{x}{a} \right), \quad a > 0$$

$$\int \frac{dx}{\sqrt{x^2 \pm a^2}} = \ln |x + \sqrt{x^2 \pm a^2}|$$

chapter (7):

Section (7.1) :- integration by parts (I.B.P)

$$\int u \, dv = uv - \int v \, du$$

$$1 - \int x^n \cdot \left(\begin{array}{c|c} \frac{\partial u}{\partial x} & \frac{dx}{dx} \\ \hline \cos x & e^x \end{array} \right) dx \rightarrow u = x^n, dv = ()$$

2- $\int x^n \cdot \left(\begin{matrix} \sin^{-1} x \\ \cos^{-1} x \\ \tan^{-1} x \end{matrix} \middle| \begin{matrix} \ln x \\ \log x \end{matrix} \right) dx \rightarrow u = (), dv = x^n$

3- $\int \begin{pmatrix} \sin x \\ \cos x \\ \tan x \end{pmatrix} \begin{pmatrix} a^x \\ e^x \\ \log x \end{pmatrix} dx \rightarrow \text{repeated integral. (من تكرار u و log u)}$
 لا سددت تكرار فالدالة لنا اي تب الحفظ على نوعي نوع الحالات.

★ غنما يكون لدينا تكامل $\int_{\gamma} f(z) dz$ حيث γ أن (ص) هي أي شيء عددا (x) نفرض أن $t = (ص)$ ومم ثم نفرض $I.B.P$.

Section (7.2) :- trigonometric integrals

$$\int \sin^n x \cos^m x \, dx \text{ or } \int \sinh^n x \cosh^m x \, dx$$

□ if (n) or (m) is odd: $\sin^2 x = 1 - \cos^2 x$, $\cos^2 x = 1 - \sin^2 x$.

② if (n) and (m) are even: $\sin^2 x = \frac{1}{2} [1 - \cos(2x)]$, $\cos^2 x = \frac{1}{2} [1 + \cos(2x)]$

$$2\sin^2\left(\frac{x}{2}\right) = 1 - \cos(x) \quad , \quad 2\cos^2\left(\frac{x}{2}\right) = 1 + \cos(x)$$

$$\sin(x) \cos(x) = \frac{1}{2} \sin(2x) \quad , \quad \sin(2x) = 2 \sin x \cos x$$

$$\cosh^2 x - \sinh^2 x = 1, \quad 1 - \tanh^2 x = \operatorname{sech}^2 x, \quad \coth^2 x - 1 = \operatorname{csch}^2 x$$

$$\sin x \cos y = \frac{1}{2} [\sin(x+y) + \sin(x-y)] \quad , \quad \cos x \cos y = \frac{1}{2} [\cos(x+y) + \cos(x-y)]$$

$$\sin x \sin y = \frac{1}{2} [\cos(x-y) - \cos(x+y)]$$

$$\cos(-x) = \cos x, \quad \sin(-x) = -\sin x, \quad \tan(-x) = -\tan x$$

$$\int \tan^n x \sec^m x \, dx \text{ or } \int k \cot^n x \csc^m x \, dx$$

□ if (n) is odd & (tan or cot) :- $\tan^2 x = \sec^2 x - 1$, $\cot^2 x = \csc^2 x - 1$

2. if (m) is even (sec or csc) :- $\sec^2 x = \tan^2 x + 1$, $\csc^2 x = \cot^2 x + 1$

2] if (n) is even and (m) is odd :- I.B.P

$$\int \cos(kx) dx = \frac{\sin(kx)}{k}, k \neq 0, \int \sin(kx) dx = \frac{-\cos(kx)}{k}, k \neq 0$$

Section (7.3) :- trigonometric substitution.

$$1 - \sqrt{a^2 - x^2} \rightarrow x = a \sin \theta$$

2. $\sqrt{a^2 + x^2} \rightarrow x = a \tan \theta$

3. $\sqrt{x^2 - a^2} \rightarrow x = a \sec \theta$

یہی اس مضمون پر
الجميع المذکور

Completing the square formula: -

$$ax^2 + bx = (cx + d)^2 + e \quad (a > 0)$$

$$c = \sqrt{a}, \quad d = \frac{b}{2\sqrt{a}}, \quad e = -\frac{b^2}{4a}$$

* كما ان يكون معامل (x^2) موجب ، إذا كان سالبو نأخذ السالب عامل مشترك .

* غالباً نقف في ! سنستخدم نظرية المربع الكامل إذا كان هناك حدود من الدرجة الثانية أو المرابعة في السطح أو النظام من المتغير .

section (7.4): - integration of rational functions by partial fractions .

assume we have $\frac{f}{g} = \frac{\text{poly}}{\text{poly}}$

1 - if $\deg(f) \geq \deg(g)$ → 1) long division .

2) Factorize .

2 - if $\deg(f) < \deg(g)$ → Factorize .

* مثل $\frac{f}{g}$ فان يكون $\frac{\text{poly}}{\text{poly}}$ ، فنأخذ substitution عام زيج الدالة ، الدالة .

* في حالة وجود جذور مفككة (كـ مارتنة الجذر مشابهة لفرم $(u = \sqrt{x})$)

Karl weierstrass substitution :-

$$u = \tan(x/2)$$

$$\cos(x) = \frac{1-u^2}{1+u^2}$$

$$dx = \frac{2du}{1+u^2}$$

$$\sin(x) = \frac{2u}{1+u^2}$$

* تستعمل غير ما

* يكون الدوال $\sin x, \cos x, \tan x \dots$

section (7.8) :- improper integrations

- integrals over ∞ interval .

[1] if $\int_a^t f(x) dx$, exists for every number $t \geq a$, then $\int_a^\infty f(x) dx = \lim_{t \rightarrow \infty} \int_a^t f(x) dx$.

[2] if $\int_t^b f(x) dx$, " " " " " $t \leq b$, then $\int_{-\infty}^b f(x) dx = \lim_{t \rightarrow -\infty} \int_t^b f(x) dx$.

→ if the limit exists it's called convergent , if it's not it's called divergent .

[3] if $\int_a^\infty f(x) dx$ and $\int_{-\infty}^b f(x) dx$ are convergent , then : $\int_{-\infty}^\infty f(x) dx = \int_a^\infty f(x) dx + \int_{-\infty}^b f(x) dx$.

- f has ∞ discont on $[a, b]$.

[1] if f is continuous on $[a, b)$ and discontinuous on b , then $\int_a^b f(x) dx = \lim_{t \rightarrow b^-} \int_a^t f(x) dx$.

[2] if f is continuous on $(a, b]$ and discontinuous on a , then $\int_a^b f(x) dx = \lim_{t \rightarrow a^+} \int_t^b f(x) dx$.

→ if the limit exists it's called convergent , if it's not it's called divergent .

[3] if f has a discontinuity at c , where $a < c < b$, and both $\int_a^c f(x) dx$ and $\int_c^b f(x) dx$ are convergent , then we define : $\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$.

Section (6.2):-

Volume

① Disks

$$V_D = \pi \int_a^b r^2 dx \text{ (or } dy)$$

② Washers

$$V_W = \pi \int_a^b (r_2^2 - r_1^2) dx \text{ (or } dy)$$

rotation axis:

① $x=a$

$$V = \int_a^b (r_2^2 - r_1^2) dy \quad \frac{dy}{dx} \quad \Big|_{x=a}$$

② $y=a$

$$V = \int_a^b (r_2^2 - r_1^2) dx \quad \frac{dx}{dy} \quad \Big|_{y=a}$$

الدالة التي يحددها محور الدوران - محور الدوران
الدالة البعيدة عن محور الدوران - محور الدوران

r_2 → محور الدوران - محور الدوران
 r_1 → محور الدوران - محور الدوران

• If a region R lies between two curves $y=f(x)$, $y=g(x)$

where $f(x) \geq g(x)$:

$$\bar{x} = \frac{1}{A} \int_a^b x [f(x) - g(x)] dx, \quad \bar{y} = \frac{1}{2A} \int_a^b [f^2(x) - g^2(x)] dx$$

$$A = \int_a^b [f(x) - g(x)] dx$$

• If a region R lies between two curves $x=h(y)$, $x=g(y)$

where $h(y) \geq g(y)$:

$$\bar{x} = \frac{1}{2A} \int_a^b [h^2(y) - g^2(y)] dy, \quad \bar{y} = \frac{1}{A} \int_a^b [h(y) - g(y)] dy$$

$$A = \int_a^b [h(y) - g(y)] dy$$

consider a system of n particles with masses $m_1, m_2, \dots, m_n$ located at the points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ in xy -plane, then:

① The origin: $M = \sum_{i=1}^n m_i x_i$

② The y -axis: $M_y = \sum_{i=1}^n m_i x_i$

③ The x -axis: $M_x = \sum_{i=1}^n m_i y_i$

④ $(\bar{x} = \frac{M_y}{m}, \bar{y} = \frac{M_x}{m})$ where $m = \sum_{i=1}^n m_i$

Section (8.1) + (8.2):- Arc length + area of surface

- If $y=f(x)$, $a \leq x \leq b$

① The length of the curve:

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

② Area of the curve if the curve is rotated:

- about x -axis: $A = \int_a^b 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$

- about y -axis: $A = \int_a^b 2\pi x \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$

- If $x=f(y)$, $a \leq y \leq b$

① The length of the curve:

$$L = \int_a^b \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

② Area of the curve if the curve is rotated:

- about x -axis: $A = \int_a^b 2\pi y \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$

- about y -axis: $A = \int_a^b 2\pi x \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$

Section (10.1) + (10.2):-

let $C: x=g(t), y=h(t), a \leq t \leq b$

• Horizontal tangent line: H.T.L

$$dy/dt = 0, \quad dx/dt \neq 0$$

• Vertical tangent line: V.T.L

$$dx/dt = 0, \quad dy/dt \neq 0$$

• length of curve:

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

• Area about:

① x -axis: $A = \int_a^b 2\pi y \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$

② y -axis: $A = \int_a^b 2\pi x \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$

Section (8.3):- moments and centers of mass:

① $y=f(x)$, $y=g(x)$ (x -axis), $a \leq x \leq b$:

$$A = \int_a^b [f(x) - g(x)] dx$$

$$\bar{x} = \frac{1}{A} \int_a^b x [f(x) - g(x)] dx, \quad \bar{y} = \frac{1}{2A} \int_a^b [f^2(x) - g^2(x)] dx$$

② $y=f(x)$, $y=g(x)$, $a \leq x \leq b$:

$$A = \int_a^b [f(x) - g(x)] dx$$

$$\bar{x} = \frac{1}{A} \int_a^b x [f(x) - g(x)] dx$$

$$\bar{y} = \frac{1}{2A} \int_a^b [f^2(x) - g^2(x)] dx$$

$$y'' = \frac{d^2 y}{dx^2} = \frac{d}{dt} \left(\frac{dy}{dx} \right)$$

$$A = \int_a^b y dx \rightarrow dx = x' = f'(t) dt$$

$$A = \int_a^b y f'(t) dt \rightarrow y = g(t), x = f(t)$$

Section (10.3):- Polar coordinates:

★ A polar curve $r=f(\theta)$ is symmetric about the polar axis if $f(-\theta)=f(\theta)$ ($r(-\theta)=r(\theta)$).

→ To convert from polar to cartesian $(r, \theta) \rightarrow (x, y)$:

$$x = r \cos \theta, \quad y = r \sin \theta$$

→ To convert from cartesian to polar $(x, y) \rightarrow (r, \theta)$:

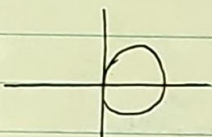
$$r^2 = x^2 + y^2, \quad \tan \theta = \frac{y}{x}$$

$\frac{\pi}{2}$	A
T	C

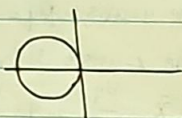
$\pi - \theta$	θ
$\pi + \theta$	$\theta - 2\pi$

$$r = \pm a \cos \theta$$

$$r = a \cos \theta$$



$$r = -a \cos \theta$$

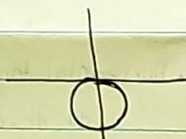


$$r = \pm a \sin \theta$$

$$r = a \sin \theta$$

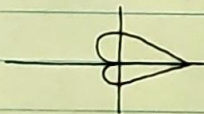


$$r = -a \sin \theta$$

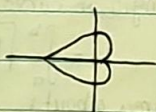


$$r = a + a \cos \theta$$

$$r = a + a \cos \theta = a(1 + \cos \theta)$$

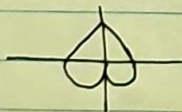


$$r = a - a \cos \theta = a(1 - \cos \theta)$$



$$r = a + a \sin \theta$$

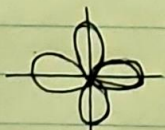
$$r = a + a \sin \theta = a(1 + \sin \theta)$$



$$r = a - a \sin \theta = a(1 - \sin \theta)$$



$$r = \cos 2\theta$$



$\cos(n\theta) = 2n$ petals, $\sin(n\theta) = n$ petals

Tangent or slopes of polar curves:

→ If $r=f(\theta)$, θ is parameter and its parametric eq's are:

$$x = r \cos \theta, \quad y = r \sin \theta$$

$$\text{then: } y' = \frac{dy/d\theta}{dx/d\theta} = \frac{dr/d\theta \sin \theta + r \cos \theta}{dr/d\theta \cos \theta - r \sin \theta}$$

Section (10.4):-

Areas and lengths of polar coordinate:

$$\textcircled{1} A = \frac{1}{2} \int_a^b [f(\theta)]^2 d\theta = \frac{1}{2} \int_a^b r^2 d\theta$$

$$\textcircled{2} A = \frac{1}{2} \int_a^b [(f(\theta))^2 - (g(\theta))^2] d\theta = \frac{1}{2} \int_a^b [r_2^2 - r_1^2] d\theta$$

$$\textcircled{3} L = \int_a^b \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

(مراجعة calculus A)

زاوية بسيطة

$$-(\sin x)' = \cos x$$

منفصلة الدالة في مشتقة الزاوية

$$-(\sin x^4)' = \cos x^4 \cdot 4x^3$$

دالة غير بسيطة

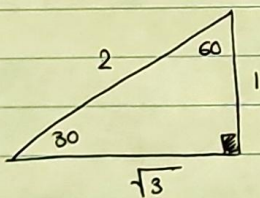
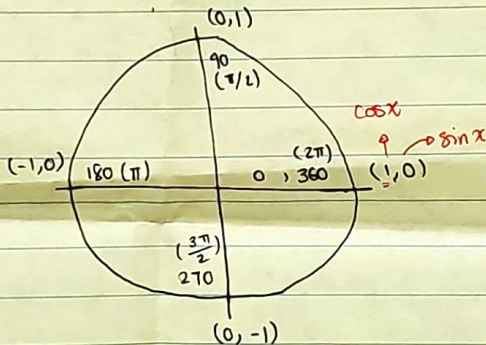
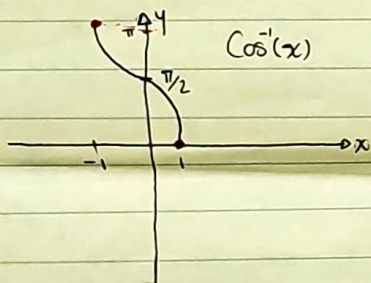
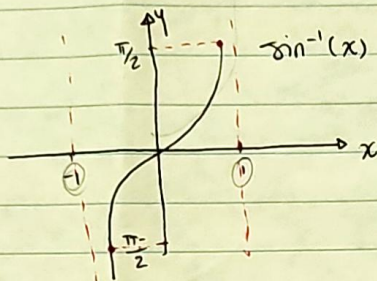
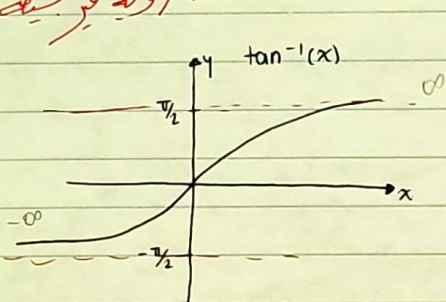
$$-(\sin^3 x)' = 3 \sin^2 x \cdot \cos x$$

دالة غير بسيطة

زاوية بسيطة

$$-(\sin^2 \sqrt{x})' = 2 \sin \sqrt{x} \cdot \cos \sqrt{x} \cdot \frac{1}{2} x^{-1/2}$$

زاوية غير بسيطة



$$\sin(45^\circ) = \cos(45^\circ) = \frac{1}{\sqrt{2}} \text{ (or) } \frac{\sqrt{2}}{2}$$

$$\tan(45^\circ) = 1$$

$$\sin \theta = \sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta$$

$$\cos \theta = \cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta$$

تتبدل الإشارة

$$\left(\int_a^b f(x) dx \right)' = 0$$

$$\left(\int_a^x f(t) dt \right)' = f(x)$$

$$\left(\int_a^{g(x)} f(t) dt \right)' = f(g(x)) \cdot g'(x)$$

$$\left(\int_{h(x)}^{g(x)} f(t) dt \right)' = f(g(x)) \cdot g'(x) - f(h(x)) \cdot h'(x)$$

Section (11.1):-

An infinite ~~inte~~ sequences: is a function whose domain is the set of positive integers.

$$a_1, a_2, a_3, \dots, a_{n-1}, a_n, a_{n+1} = \{a_n\}_{n=1}^{\infty}$$

n factorial (n!):

$$n! = n \times (n-1) \times (n-2) \times (n-3) \dots \times 2 \times 1$$

$$0! = 1, \quad \infty! = \infty$$

Convergence of sequences:-

1) If $\lim_{n \rightarrow \infty} a_n = L$ (exist) we say that $\{a_n\}$ converge.

2) If $\lim_{n \rightarrow \infty} a_n = (\text{DNE})$ fails to exist we say that $\{a_n\}$ diverge.

Theorem: If $\lim_{n \rightarrow \infty} |a_n| = 0$ then $\lim_{n \rightarrow \infty} a_n = 0$.

Sandwich theorem:

let $\{a_n\}, \{b_n\}, \{c_n\}$ be a sequences of real numbers. If $\{a_n\} \leq \{b_n\} \leq \{c_n\}$ for all n , and if

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} c_n = L$$

then, $\lim_{n \rightarrow \infty} b_n = L$

Theorem: let $\{a_n\}$ and $\{b_n\}$ be a sequences of real number and let A and B be real numbers. If

$$\lim_{n \rightarrow \infty} a_n = A \text{ and } \lim_{n \rightarrow \infty} b_n = B \text{ then,}$$

- $\lim_{n \rightarrow \infty} (a_n \pm b_n) = A \pm B$
- $\lim_{n \rightarrow \infty} (a_n \cdot b_n) = AB$
- $\lim_{n \rightarrow \infty} (k \cdot b_n) = kB$ (k is any real number).
- $\lim_{n \rightarrow \infty} a_n/b_n = A/B, \quad B \neq 0$

Section (11.2):-

Infinite series:- the sum of infinite seq. conv. of series:

$$a_n = s_n - s_{n-1}$$

$$\lim_{n \rightarrow \infty} s_n = L \rightarrow \{s_n\} \text{ conv.} \Rightarrow \sum_{n=1}^{\infty} a_n = L \text{ conv.}$$

$$\lim_{n \rightarrow \infty} s_n = (\text{DNE}) \rightarrow \{s_n\} \text{ div.} \Rightarrow \sum_{n=1}^{\infty} a_n \text{ div. (no sum)}$$

Telescoping series (T.S):-

$$\rightarrow () - ()$$

$$\rightarrow \frac{()}{()} \text{ P.F.} \rightarrow () - ()$$

Geometric series (G.S):-

$$\sum_{n=1}^{\infty} ar^{n-1} = a + ar + ar^2 + ar^3 + \dots + ar^{n-1} + \dots$$

- a is the first term, $a \neq 0$.

- r is the ratio.

$\rightarrow$ The G.S $\sum_{n=1}^{\infty} ar^{n-1}$ is conv. If:

$$|r| < 1, \quad \text{Sum} = \frac{a}{1-r}$$

$\rightarrow$ The G.S $\sum_{n=1}^{\infty} ar^{n-1}$ is div. If: $|r| > 1$, no sum.

$\rightarrow$ أوفى في الـ (G.S) عندما يكون $|r| < 1$ (عدد) أو $|r| > 1$ (عدد) أو $|r| = 1$ (عدد) أو $|r| < 1$ (عدد) أو $|r| > 1$ (عدد) أو $|r| = 1$ (عدد).

2) The series $\sum_{n=1}^{\infty} \frac{1}{n}$ is called harmonic series (H.S). It is always div.

4) The series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ is called p-series. It's conv if $p > 1$, and div if $p \leq 1$.

5) (n-th term test for divergens):

If $\lim_{n \rightarrow \infty} a_n \neq 0$, then $\sum_{n=1}^{\infty} a_n$ is Div.

$\rightarrow$ نفق غالباً في الاستخدام هذا الاختبار عندما تكون النهاية على شكل $(\frac{\infty}{\infty})$.

Theorem:-

If $\sum a_n = A$ and $\sum b_n = B$ are convergent series.

$$\rightarrow \sum a_n \pm \sum b_n = A \pm B \text{ is conv.}$$

If k is non-zero constant then $\sum k a_n = kA$ is conv.

$$\text{Conv} \pm \text{Conv} = \text{Conv.}$$

$$\text{Conv} \pm \text{Div} = \text{Div.}$$

$$\text{Div} \pm \text{Div} = \text{may be Conv or div.}$$

Section (11.3):-

Integral test: let $\sum_{n=1}^{\infty} a_n$ be a series with positive terms. and let $f(x)$ be a function obtained by replacing n by x in the formula of a_n , if:

1) $f(x)$ is positive.

2) $f(x)$ is cont. on $[N, \infty)$.

3) $f(x)$ is decreasing For all $n \geq N$.

Then the series $\sum_{n=1}^{\infty} a_n$ and $\int_N^{\infty} f(x) dx$ are both either conv. or div.

if $\heartsuit$ is conv. $\rightarrow$ $\heartsuit$ Conv.

if $\heartsuit$ is div. $\rightarrow$ $\heartsuit$ div.

• أنف غالباً في هذا الاختبار • لا نعرف نكامل الدالة بصورة
• لو قمنا بـ $u = (n)$ ووجدت المشتقة أو جزء
• منها بالسؤال.

Section (11.4):-

Comparison test and limit comparison test.

→ The Comparison test (C.T):

Let $\sum a_n$, $\sum c_n$ and $\sum b_n$ be series with positive terms, suppose that for some integer N :

$$\begin{array}{ccc} d_n & \leq & a_n \leq c_n \\ \downarrow & \nearrow & \nwarrow \\ \text{Div} & & \text{Conv} \end{array}$$

- If $\sum c_n$ is conv., then $\sum a_n$ is also conv.

- If $\sum d_n$ is div., then $\sum a_n$ is also div.

في (C.T) :-

1- إذا اتجهت يميناً: نكتب البـ $\heartsuit$ تمثيل المقام

2- إذا اتجهت يساراً: $\heartsuit$ تمثيل البـ $\heartsuit$ نكتب المقام

• غالباً إذا رأينا: $\sin(n)$ أو $\cos(n)$ نفك بال (C.T)

أو إذا رأينا: $(n!)$ نفك في (R.T) أو (C.T)

→ The limit Comparison test (L.C.T):

suppose that $a_n > 0$ and $b_n > 0$, $\forall n \geq N$.

① If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = C > 0$, then $\sum a_n$ and $\sum b_n$ are both conv. or both div.

② If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 0$ and $\sum b_n$ Conv., then $\sum a_n$ Conv.

③ If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \infty$ and $\sum b_n$ div., then $\sum a_n$ div.

• غالباً اختيار الـ (b_n) : أي في البـ

• أي في المقام $\heartsuit$ تستعمل هذا

الاختبار. كاستخدم مع $\sin(n)$ أو $\cos(n)$ $(n!)$

Section (11.6):-

(ratio and root test)

→ The ratio test (R.T):-

Let $\sum a_n$ be a series with +ve terms ($a_n \geq 0, \forall n \geq N$).

And suppose that $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \rho$. Then:-

① The series $\sum a_n$ Conv. if $\rho < 1$.

② The series $\sum a_n$ div. if $\rho > 1$.

• the test is inconclusive (fail) if $\rho = 1$.

→ The n-th root test:-

Let $\sum a_n$ be a series with +ve terms ($a_n \geq 0, \forall n \geq N$).

And suppose that $\lim_{n \rightarrow \infty} \sqrt[n]{a_n} = \rho$.

1- The series $\sum a_n$ Conv. if $\rho < 1$.

2- The series $\sum a_n$ div. if $\rho > 1$.

• the test is 'conclusive (fail)' if $\rho = 1$.

Section (11.5):-

(Alternating series test):

$$\sum_{n=1}^{\infty} (-1)^{n+1} a_n = a_1 - a_2 + a_3 - a_4 + \dots + (-1)^{n+1} a_n + \dots$$

$$\text{(Or)} \sum_{n=1}^{\infty} (-1)^n a_n = -a_1 + a_2 - a_3 + a_4 + \dots + (-1)^n a_n + \dots$$

→ The Alternating series test (A.S.T):-

the alternating series $\sum_{n=1}^{\infty} (-1)^{n+1} a_n$ Conv. if the

following three conditions are satisfied:

① $\lim_{n \rightarrow \infty} a_n = 0$ ② the a_n 's are all positive.

③ the positive a_n 's are decreasing, that is

$a_n \geq a_{n+1}, \forall n \geq N$ for some integral N .

Conv $\rightarrow$ A.C.

Div $\left\{ \begin{array}{l} \lim_{n \rightarrow \infty} a_n \neq 0 \rightarrow \text{div} \\ \text{3 conditions satisfied} \rightarrow \text{C.C} \end{array} \right.$

Alternating series estimation:

If $S = \sum_{n=1}^{\infty} (-1)^{n+1} b_n$ is the sum of alternating series that satisfies: ① b_n 's are all +ve.

② $b_n \geq b_{n+1}, \forall n \geq N$.

③ $\lim_{n \rightarrow \infty} b_n = 0$

Then $|R_n| = |S - S_n| \leq b_{n+1}$

→ this theorem for conv alternating series only!!

Section (11.8):

The power series:-

$$\sum_{n=0}^{\infty} a_n (x-c)^n \text{ or } \sum_{n=0}^{\infty} a_n x^n, x \neq 0$$

* تعامل ال (x) كتابت

$$\frac{b-a}{2} = \text{radius of conv.} \quad \text{هنا! يبار}$$

interval of Conv.

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| \quad \text{ratio test} \quad \text{لا يبار! لا نستطيع}$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} \quad n\text{-th root test}$$

$$-R < x < R \quad \text{سوف يظهر لنا}$$

$$\text{لازم ندرس ال (x) عند الاقتراف}$$

$$\text{Div كانت} \quad \text{فوسر مفتوح} \quad (a, b)$$

$$\text{Conv كانت} \quad \text{فوسر مغلق}$$

Power series summary

$$\textcircled{1} \text{ if } x \in I = [a, b], (a, b), (a, b], [a, b]$$

$$R = \frac{b-a}{2}, \quad \lim_{n \rightarrow \infty} = 1$$

$$\textcircled{2} \text{ if } x = I = a$$

$$R = 0, \quad \lim_{n \rightarrow \infty} = \infty$$

$$\textcircled{3} \text{ if } x \in (-\infty, \infty)$$

$$R = \infty, \quad \lim_{n \rightarrow \infty} = 0$$

Section (11.9):

recall:

$$1 - \sin x \cos x = \frac{1}{2} \sin(2x)$$

$$\sin(2x) = 2 \sin x \cos x$$

$$2 - \cos^2 x = 1/2 (1 + \cos 2x)$$

$$3 - \sin^2 x = 1/2 (1 - \cos 2x)$$

$$4 - \cosh^2 x = 1/2 (\cosh(2x) + 1)$$

$$5 - \sinh^2 x = 1/2 (\cosh(2x) - 1)$$

$$6 - \sinh x \cosh x = 1/2 \sinh(2x)$$

$$7 - a^n = e^{n \ln a}$$

Section (11.10):

Taylor and Maclaurin series:

→ The Taylor series generated by the function f at the point a is:

$$P(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)(x-a)^n}{n!}$$

$$= f(a) + \frac{f'(a)(x-a)^1}{1!} + \frac{f''(a)(x-a)^2}{2!} + \dots + \frac{f^{(n)}(a)(x-a)^n}{n!}$$

→ If $(a=0)$ we get the Maclaurin series that is:

$$P(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)(x)^n}{n!}$$

$$= f(0) + \frac{f'(0)x}{1!} + \frac{f''(0)x^2}{2!} + \dots + \frac{f^{(n)}(0)x^n}{n!}$$

remark:-

$$1 - \sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$$

$$2 - \cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$$

$$3 - 1, -1, 1, -1, \dots, (-1)^{n+1}, \dots$$

$$4 - -1, 1, -1, 1, \dots, (-1)^n, \dots$$

$$5 - 0, 2, 4, 6, 8, \dots, 2n, \dots$$

$$6 - 1, 3, 5, 7, 9, \dots, 2n+1, \dots$$

$$7 - (n) \text{ degree Taylor polynomial of } f(x) \text{ at } x=a.$$

$$T_n(x) = f(a) + \frac{f'(a)(x-a)}{1!} + \frac{f''(a)(x-a)^2}{2!} + \dots + \frac{f^{(n)}(a)(x-a)^n}{n!}$$

→ The binomial series:

If (k) is any real number and $|x| < 1$, then:

$$(1+x)^k = \sum_{n=0}^{\infty} \binom{k}{n} x^n$$

$$= 1 + kx + \frac{k(k-1)}{2!} x^2 + \frac{k(k-1)(k-2)}{3!} x^3$$

$$\text{where } \binom{k}{n} = \frac{k(k-1)(k-2) \dots (k-n+1)}{n!}$$

section (7.2) :-

$$- \int \sin^n x \cos^m x dx$$

[1] if (n) or (m) is odd :-

$$\sin^2 x = 1 - \cos^2 x, \cos^2 x = 1 - \sin^2 x$$

[2] if (n) and (m) are even :-

$$\sin^2 x = \frac{1}{2} [1 - \cos(2x)]$$

$$\cos^2 x = \frac{1}{2} [1 + \cos(2x)]$$

$$\left[\begin{aligned} 2\sin^2\left(\frac{\theta}{2}\right) &= 1 - \cos(\theta) \\ 2\cos^2\left(\frac{\theta}{2}\right) &= 1 + \cos(\theta) \end{aligned} \right]$$

$$\sin(x) \cos(x) = \frac{1}{2} \sin(2x)$$

$$\cosh^2 x - \sinh^2 x = 1$$

$$1 - \tanh^2 x = \operatorname{sech}^2 x$$

$$\coth^2 x - 1 = \operatorname{csch}^2 x$$

$$\sin x \cos y = \frac{1}{2} [\sin(x+y) + \sin(x-y)]$$

$$\cos x \cos y = \frac{1}{2} [\cos(x+y) + \cos(x-y)]$$

$$\sin x \sin y = \frac{1}{2} [\cos(x-y) - \cos(x+y)]$$

$$\cos(-x) = \cos x, \sin(-x) = -\sin x$$

$$\tan(-x) = -\tan x$$

$$\int \tan^n x \sec^m x dx \text{ (or) } \int \cot^n x \csc^m x dx$$

1] If (n) is odd (tan or cot) :-

$$\tan^2 x = \sec^2 x - 1$$

$$\cot^2 x = \csc^2 x - 1$$

2] If (m) is even (sec or csc) :-

$$\sec^2 x = \tan^2 x + 1$$

$$\csc^2 x = \cot^2 x + 1$$

3] If (n) is even and (m) is odd :-

I.B.P

$$\sinh^2 x = \cosh^2 x - 1$$

$$\sin(2\theta) = 2 \sin \theta \cos \theta$$

$$\cos(2\theta) = \cos^2 \theta - \sin^2 \theta$$

section (11.9):-

Representation of functions as power series in power of x .

1- $\frac{1}{1+u}$ $\sum_{n=0}^{\infty} (-1)^n u^n$ $|u| < 1$ $I = (-1, 1)$ $R = 1$

2- $\frac{1}{1-u}$ $\sum_{n=0}^{\infty} u^n$ $|u| < 1$ $I = (-1, 1)$ $R = 1$

3- $\ln(1+u)$ $\sum_{n=0}^{\infty} (-1)^n \frac{u^{n+1}}{n+1}$ $I = (-1, 1]$ $R = 1$

4- $\tan^{-1} u$ $\sum_{n=0}^{\infty} (-1)^n \frac{u^{2n+1}}{2n+1}$ $I = [-1, 1]$ $R = 1$

5- e^u $\sum_{n=0}^{\infty} \frac{u^n}{n!}$ $I = \mathbb{R}$ $R = \infty$

6- $\sin u$ $\sum_{n=0}^{\infty} (-1)^n \frac{u^{2n+1}}{(2n+1)!}$ $I = \mathbb{R}$ $R = \infty$

7- $\sinh u$ $\sum_{n=0}^{\infty} \frac{u^{2n+1}}{(2n+1)!}$ $I = \mathbb{R}$ $R = \infty$

8- $\cos u$ $\sum_{n=0}^{\infty} (-1)^n \frac{u^{2n}}{(2n)!}$ $I = \mathbb{R}$ $R = \infty$

9- $\cosh u$ $\sum_{n=0}^{\infty} \frac{u^{2n}}{(2n)!}$ $I = \mathbb{R}$ $R = \infty$

Basic limits:

1- $\lim_{n \rightarrow \infty} \frac{\ln(n)}{n} = 0$

2- $\lim_{n \rightarrow \infty} \sqrt[n]{n} = 1$

3- $\lim_{n \rightarrow \infty} (x)^{1/n} = 1 \quad x > 0$

4- $\lim_{n \rightarrow \infty} (x)^n = \begin{cases} 0 & -1 < x < 1 \\ 1 & x = 1 \\ \infty & x > 1 \\ \text{DNE} & x \leq -1 \end{cases}$

5- $\lim_{n \rightarrow \infty} \frac{(x)^n}{n!} = 0 \quad \text{For any } x.$

6- $\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x \quad \text{For any } x.$

7- $\lim_{n \rightarrow \infty} \frac{\sin(k/n)}{(k/n)} = 1 \quad k \in \mathbb{R}.$

8- $\lim_{n \rightarrow \infty} \frac{\tan(k/n)}{(k/n)} = 1$

9- $\lim_{n \rightarrow \infty} \frac{1 - \cos(k/n)}{(k/n)} = 0$

Section (11.2):-

$$S_n = a_1 + a_2 + a_3 + \dots + a_{n-1} + a_n$$

$$S_{n-1} = a_1 + a_2 + a_3 + \dots + a_{n-1}$$

$$S_n = S_{n-1} + a_n$$

$$a_n = S_n - S_{n-1}$$

$$- \lim_{n \rightarrow \infty} S_n = L \rightarrow \{S_n\} \text{ Conv. } \Rightarrow \sum_{n=1}^{\infty} a_n = L \text{ Conv.}$$

$$- \lim_{n \rightarrow \infty} S_n = (\text{DNE}) \rightarrow \{S_n\} \text{ Div. } \Rightarrow \sum_{n=1}^{\infty} a_n \text{ no sum Div.}$$

(الفانكشن)

① Telescoping series (T.S.):

$$1 - () - () \quad 2 - \frac{()}{()} \rightarrow P.F \rightarrow () - ()$$

② Geometric series (G.S.):

$$\sum_{n=1}^{\infty} ar^{n-1} = a + ar + ar^2 + ar^3 + \dots + ar^{n-1}$$

- a is the first term, $a \neq 0$.

- r is the ratio.

→ The G.S. $\sum_{n=1}^{\infty} ar^{n-1}$ is Conv. If $|r| < 1$, sum = $\frac{a}{1-r}$

→ The G.S. $\sum_{n=1}^{\infty} ar^{n-1}$ is Div. If $|r| \geq 1$, no sum

* أفكر في الـ G.S. عندما يكون لي r^n (عدد)، أو أزل نوعها بعد n ،
أو عندما تكون قسمت أي عدد على سابقته مقدار ثابت.

③ Harmonic series (H.S.):

$$\sum_{n=1}^{\infty} \frac{1}{n}, \text{ It is Div always.}$$

④ p-series:-

$$\sum_{n=1}^{\infty} \frac{1}{n^p}, \text{ Conv. If } p > 1 \text{ and Div. If } p \leq 1.$$

— If the series $\sum_{n=1}^{\infty} a_n$ Conv. the limit $a_n = 0$ and the Converse is not true.

⑤ n-th term test for diverges:-

If $\lim_{n \rightarrow \infty} a_n \neq 0$, then $\sum_{n=1}^{\infty} a_n$ is div.

→ If it equal zero we need to use another test.

section (11.3):- (Integral test):-

Let $\sum_{n=1}^{\infty} a_n$ be a series with positive terms, and let $f(x)$ be a function obtained by replacing (n) by (x) in the formula of a_n if:

① $f(x)$ is positive. ② $f(x)$ is cont. on $[N, \infty)$.

③ $f(x)$ is decreasing for all $n \geq N$.

Then, the series $\sum_{n=1}^{\infty} a_n$ and $\int_N^{\infty} f(x) dx$ are both Conv. or div.

if ♥ is Conv. → the ☼ will be Conv.

if ♥ is div. → the ☼ will be div.

أنتي غالباً في هذا الاختبار إذا: ① لو تعرفي زكامة الدالة بكل سهولة.

② لو غمزت ————— (٥٥) = ٤ ووجدت المشتقة أو غير ذلك منها

في السؤال.

Section (11.4):-

- The Comparison test:

let $\sum A_n$, $\sum C_n$, and $\sum d_n$ be series with positive terms.

$$d_n \leq A_n \leq C_n$$

div

Conv.

- في ال (C.T):

١- إذا اتجهت $\sum C_n$ بمبدأ: $\sum C_n$ يتبع المقام $\frac{1}{n!}$ < تصغير المقام.
٢- إذا اتجهت $\sum d_n$ بمبدأ: $\sum d_n$ يتبع المقام $\frac{1}{2^n}$ < تصغير المقام.

① If $\sum C_n$ is conv., then $\sum A_n$ also conv.

② If $\sum d_n$ is div., then $\sum A_n$ also div.

$$2^n > n \quad \frac{1}{n!} < \frac{1}{2^n - 1}$$

غالبًا إذا رأينا $\sin(n)$ أو $\cos(n)$ نفكر باستخدم ال (C.T.) وإذا رأينا $n!$ نستعمل ال (C.T.) أو ال (R.T.).

- The limit Comparison test:

Suppose that $a_n > 0$ and $b_n > 0$, $\forall n \geq N$

① If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = C > 0$, then $\sum a_n$ and $\sum b_n$ are both Conv. or div.

② If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 0$ and $\sum b_n$ Conv., then $\sum a_n$ Conv.

③ If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \infty$ and $\sum b_n$ div., then $\sum a_n$ div.

★ لا اختيار ال (b_n) = أكبر أو أقوى / أصغر أو أقوى / أضعف أو أقوى في المقام

غالبًا إذا رأينا $\sin(1/n)$ أو $\cos(1/n)$ نستعمل هذا الاختيار.

★★ نستعمله مع $\sin(n)$ < $\cos(n)$ أو $n!$.

Section (11.6):-

- The ratio test (R.T):-

Let $\sum a_n$ be series with +ve terms ($a_n \geq 0, \forall n \geq N$). Suppose that $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \rho$, then:

1- The series is Conv. if $\rho < 1$.

2- The series is div. if $\rho > 1$.

→ The test is inconclusive (fail) if $\rho = 1$.

- The n-th root test:-

Let $\sum a_n$ be series with +ve terms ($a_n \geq 0, \forall n \geq N$). Suppose that $\lim_{n \rightarrow \infty} \sqrt[n]{a_n} = \rho$, then:-

1- The series $\sum a_n$ is Conv. if $\rho < 1$.

2- The series $\sum a_n$ is div. if $\rho > 1$.

→ The test is inconclusive (fail) if $\rho = 1$.